

OCTOBER, 2026

DECODING THE RHYTHMS OF EMOTION: A SENTIMENTAL JOURNEY THROUGH MUSIC GENRES

GROUP PROJECT
TEXT MINING 2026/2027
DUE: DECEMBER 22ND (17:59)



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I. INTRODUCTION

In the vast world of music, lyrics often serve as a mirror, reflecting the emotions, thoughts, and narratives of the artist. These lyrical themes can vary significantly across different music genres, painting a diverse emotional landscape. This project aims to explore this landscape through the lens of Natural Language Processing (NLP).

II. PROJECT GOALS

The goal of your project is two-fold:

1. **Genre Identification:** Create a classification model that can accurately predict the genre of a song from its lyrics. This will involve training a machine learning model on a dataset of song lyrics and their corresponding genres.
2. **Sentiment Analysis:** Examine the emotional undertones within song lyrics to decipher the emotional essence of each genre. You'll need to derive sentiment ratings from the lyrics and look over these ratings to find out about notable sentiment differences within and across genres. This segment of the project is open-ended, meaning that you can formulate and answer as many questions as you desire. **Here are some thought-provoking questions you may consider addressing in your sentiment analysis:**
 - a. What are the predominant sentiments in the lyrics of the songs of a specific musical genre?
 - b. Are there changes in the predominant sentiments of a genre over the years?
 - c. Does a song's sentiment impact its popularity within a genre (or even across genres)?

III. DATASET

You have access to two different datasets:

In the **training set**, you will find the features and the ground truth associated with each song, i.e. the musical genre. Use it to build your machine-learning models. The goal will be to use the model you created and make predictions on unseen data (i.e. your test set).

In the test set, you will still have access to the song lyrics and other descriptive attributes of each song. However, you will not have access to the musical genre. In the text classification portion, your goal will be to predict the musical genre from the **song lyrics**.

The available data contains the following attributes:

ATTRIBUTE	DESCRIPTION
id	Unique identifier
title	Song Title
artist	The main artist of the song
year	Year the song was launched in
views	Number of views the song has on an online music service
features	Includes names of collaborating artists, empty otherwise
lyrics	The lyrics of the song. Includes tags for specific sections (e.g. Introduction, Chorus).
tag	The main musical genre of the song

IV. OUTLINE

Your project deliverables should respect the following outline:

Abstract

A small summary of your work (200 to 300 words). The abstract should give an overview of your work: What is the context? What is your main hypothesis? What did you do? What were your main results, and what conclusions did you draw from them?

I. Introduction

- Overview of the project
- Main goals of the project (being a requirement for the course does not count)
- Did you find any research with similar objectives? What has been done? What did other researchers find? What would you expect your results to be based on their previous findings?

II. Data Exploration and Preprocessing

- Description of data received
- Steps taken to clean and prepare the data

III. Genre Identification

- Additional preprocessing steps adopted
- Description of the actions taken
- Results and discussion of main findings

IV. Sentiment Analysis

- Explanation of the process
- Results and findings

V. Conclusion

- Summary of the findings
- Do the findings match what you initially expected? How?
- Discussion of limitations of your work (e.g. what could you have done differently)
- Suggestions for possible work to follow your work.

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V. DELIVERABLES

Upon the project's deadline, you will be required to submit:

- A Jupyter notebook (or a zip of multiple notebooks) featuring all the code you used throughout the project to:
 - a. Decide on your final solution
 - b. Obtain your final results (code that helped you make decisions but does directly contribute to reach should be included, but commented.
 - c. Any Markdown Comments useful to guide us through the project rationale.

VI. EVALUATION

Your work will be evaluated according to the following criteria:

CRITERIA	PERCENTAGE (%)	MAXIMUM GRADE (OUT OF 20)
Notebook Quality	15	3
Data Exploration	10	2
Initial data preprocessing	20	4
Text Classification	20	4
Sentiment Analysis	20	4
Conclusion	10	2
Creativity and Other Self-studies	5	1

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Your grade will reflect our assessment of the quality of your work in terms of quality of writing, clarity, conciseness, correctness and efficiency. Please find below more details about what is taken into account for each topic:

- **Notebook Quality and Storytelling (3v):** A Your notebook(s) should be readable and understandable to someone reading and looking at them for the first time. Every cell you present should have a clear purpose. This section also encompasses the overall quality of your introduction and conclusions for each cell.
- **Data Exploration (2v):** Describe the data and extract meaningful insights that you consider helpful. Avoid adding visualizations and elements that add nothing to address the problem at hand.
- **Initial data preprocessing (4v):** This section covers the initial preprocessing of your data. In essence, it should unambiguously explain the steps and rationale behind your conversion of text into numbers: what form/forms of representation did you consider and why? Did you remove stopwords? If you did, why and if you didn't, also why? Was the data normalized or scaled in any way? Did you use the entire song as the full corpus or split different song sections into different columns?
- **Text Classification (4v):** Describe your strategy for the text classification objective. This section is separated into different components:
 - Kaggle Performance: 0.5v
 - Additional Preprocessing (includes feature selection): 1v
 - Modelling approach (model assessment (holdout, cross-validation, etc...), algorithms used): 1.5v
 - Performance assessment (metrics and interpretation of results): 1v
- **Sentiment Analysis (4v):** Describe your strategy for the text classification objective. This section is more open-ended and is separated into the following components:
 - Adequacy of your goals for the sentiment analysis: 0.5v
 - Difficulty of the tasks you are performing: 1v
 - Correctness of your implementation: 1.5v
 - Alignment between your goals and your results: 0.5v
- **Creativity and other self-studies (1v):** This topic includes applying different techniques and aspects of creativity, such as choice of visualizations, approach or techniques used. If techniques not given during practical classes are used, you should provide a theoretical explanation for them in the annexes.

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VII. PARTING NOTES

1. Deliveries made after the deadline will incur a penalty of 1 point per day. Deliveries made before the deadline will receive a bonus of 0.15 points per day of delivery in advance (up to a maximum of 1 point).
2. If you use automated packages (e.g., ydata, pandas-profiling), we will only consider observations that have a visible impact on the project (i.e., a key insight that is acted upon).
3. When it comes to creating Predictive Models, you are allowed to create your own implementations or use one of the following packages: vanilla Scikit-Learn, Tensorflow, Keras, PyTorch (and skorch) or Transformers.
4. Using Lazy Predict or similar AutoML (e.g., Feature Tools, TSFresh) packages, XGBoost, CatBoost, LightGBM (or similar packages not in vanilla sk-learn) is explicitly off-limits and will result in a 1-point penalty per attempt.
5. We won't consider any steps or results that are not mentioned in any of your submitted notebooks. Everything in your clean notebook must have a clear purpose. Avoid including irrelevant, unimportant, or redundant information. Additionally, please refrain from providing theoretical explanations of topics covered in class.
6. The trustworthiness of the information you provide is key. You should look to source information from peer-reviewed journals (thus, avoid citing Medium, TowardsDataScience and similar sources). Also, for the love of God, DO NOT CITE ChatGPT or any other LLM for that matter.
7. Before submitting, run your notebook from the start one last time (if you used a GridSearch, you can comment this cell, but you should run the final model with the GS parameters in a different cell).
8. If you tested more elements that should be considered, but did not feature in your clean notebook, please submit the code you used in a separate submission form created for that purpose.
9. We will run your Jupyter Notebooks. Please ensure that we can run the notebook from start to finish without interruption. Notebooks that do not fulfil this condition will be severely penalized.
10. The code will pass through a process of plagiarism and AI generation checking.
11. When determining the grade for your work, there will be a comparative component between it and the work presented by your peers.

Friendly Reminders:

1. The order in which the group members present their notebooks is decided by the professors at the start of the discussion. Not attending will result in a 0.
2. If something is good enough to be mentioned, it is also good enough to know. DO NOT include techniques/algorithms/steps you cannot explain in your work: we may (and probably will) ask about them in the discussion.
3. Finished is better than perfect.